Xnergy

Autonomous Power Technologies

ORIGIN FOR (SG) WESTON ROBOT

Application Note

BEAC018- Electronics Clearance and Shielding Guidelines

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Abstract

This article provides guidelines for clearance and shielding of electronics devices surrounding transmitter pad and receiver charging unit from unwanted influences of magnetic flux. This is to ensure proper operations of the surrounding electronics. This allow engineers to quickly and efficiently design and test their system in vicinity of Xnergy BE series wireless charger.

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1. Introduction

Xnergy BE series wireless charging charges electrical driven applications through the medium of air without physical contact. The wireless charging process uses magnetic field that allows energy to be transferred through magnetic induction. It consists of the transmitter powering unit (TPU) with transmitter pad (TXP) and compact receiver charging unit (RCU) as shown in Figure 1. In the applications, RCU is mounted on mobile robotics to receive the wireless power and charge the battery in the robot. The TPU and TXP are normally mounted on stationery stand or on the floor.

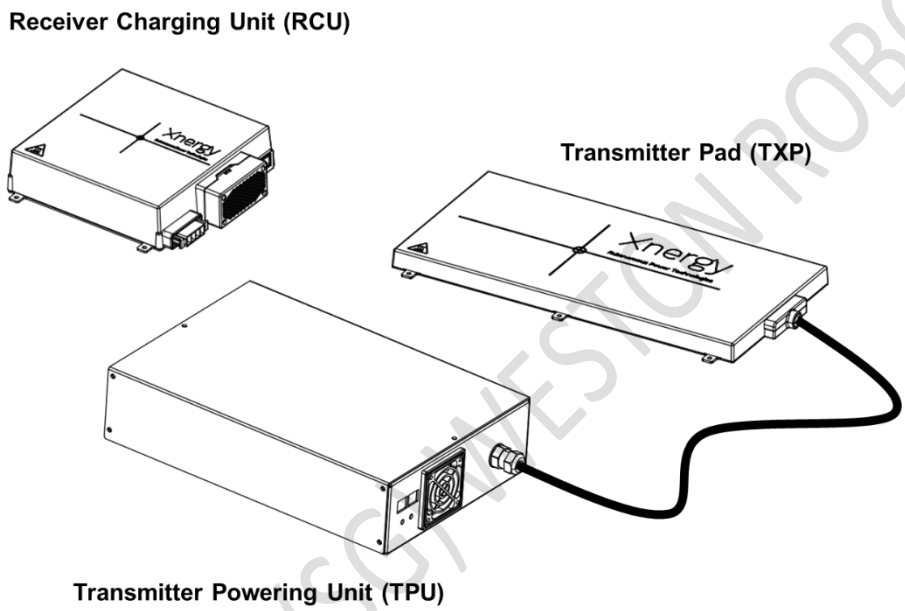

Figure 1: Xnergy BE series wireless charger

The magnetic field linking the TXP and RCU form continuous loop which extends outside the coil for the return path as shown in Figure 2. These magnetic field paths need to be considered in the installation of wireless charger. They might heat up conductive, metallic, or soft magnetic objects, or they might interfere with the operations of electronics parts that are sensitive to magnetic field. Therefore, electronics clearance distance and shielding need to be adhered in the installation.

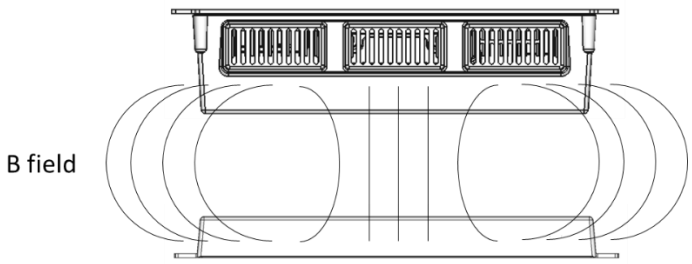

Figure 2: Continuous magnetic field lines between transmitter and receiver

This article focuses on the electronics at the surrounding of the TXP and RCU, such as the electronics in the mobile robotics. Due to space constraints in the mobile robotics, passive parts and electronics maybe packed close to each other.

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2. Electromagnetic immunity of electronics devices

Wireless charger is commonly used in robots such as autonomous guided vehicle (AGV) and autonomous mobile robot (AMR). While their features and navigation system are different, they have similar basic structure such as battery, traction system, navigational sensors, obstacle sensors, controllers, etc. [1]. In other robots, electronics devices might include RFID sensor, magnetic sensor, QR code reader, laser sensor, line tracer, LIDAR, etc.

These electronics operate differently under exposure to magnetic field of various frequency and magnitude. While some of them pass a certain immunity standard, some might not.

Example of some common sensors in mobile robotics and their EMC standard are shown in Table 1.

Table 1: Example of sensors and their EMC standards

Item	Part	EMC standard
RFID sensor	SICK RFH620-1000001	EN 301 489-3
QR code reader	SICK V2D621R-2MSFFB5	EN 61000-6-2:2005-08 / EN 61000-6-4 (2007-01) + A1 (2011)
Magnetic line guidance sensor	SICK MLSE-0600A2TP0	EN 60947-5-2

A commonly used Electromagnetic compatibility (EMC) standards EN 61000-6-2: Generic standards-Immunity for industrial environments covers only specific frequency of 50 Hz, 60 Hz, 150 kHz – 80 MHz, and above.

Therefore there is a gap at low frequency range between 60 Hz and 150 kHz. This happens to be the frequency where most power electronics products operate, including wireless charger.

3. Electronics shielding guidelines.

Varieties of electronics that have different working principles, different immunity to stray field, and different EMC standards present challenges in protecting them against the magnetic flux used in wireless charger.

“In the absence of standards specifying compatibility levels, immunity requirements or emission limits, product standards did not consider the frequency range 2 kHz – 150 kHz, as there could not be referenced to related EMC standards” [2].

Therefore the best protection for these electronics are:

- Large clearance distance from wireless charger system (TXP and RCU)
  This method may not be practical especially mobile robotics are compact.
- Shielding from stray flux generated by wireless charger system

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This method is preferred for compact applications, where electronics can be close to wireless charger system if the heating in the shielding is acceptable.

Shielding of electronics works by providing conductive material between the protected electronics at certain distance from RCU and TXP. This conductive material shields the electronics by countering the stray magnetic field based on the Eddy current principle but there is heating effect in the conductive material.

The following sections give the guidelines for shielding materials and distance from RCU and TXP.

Shielding materials

The shielding must be made by conductive non-ferrous materials. Aluminium with thickness ≥ 1 mm is recommended. Avoid steel and iron.

A full Aluminium sheet is ideal for shielding. However, holes for air ventilations maybe needed in some cases, for example when RCU is fully enclosed. In this case, perforated Aluminium plate with holes of diameter ≤ 3 mm can be used. A sample of such perforated Al plate is RS Pro 202-7912 as shown in Figure 3.

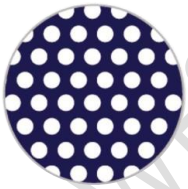

Figure 3: Perforated metal

Shielding distance to RCU

The shielding distance guidelines are set for different RCU mounting to the robot. These guidelines are set based on in house testing for acceptable temperature increase in shielding material (less than 10 degree Celsius) .

1. Protruding exposed RCU
   RCU is mounted with protruding surface out of the robot as shown in Figure 4. The electronics can be installed in the shielded area (grey coloured area) behind the metal sheet.

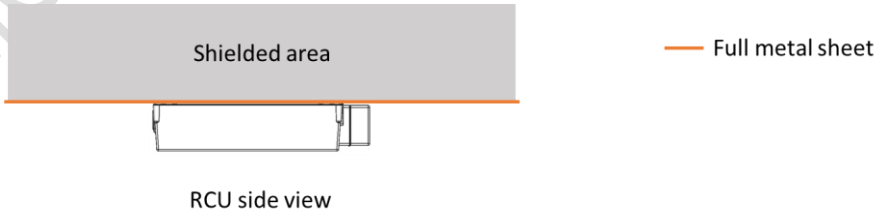

Figure 4: Shielding distance for protruding exposed RCU mounting

2. Flushed exposed RCU
   RCU is mounted with flushed surface with the robot as shown in Figure 5. The RCU is not enclosed completely, therefore allowing air flow at surrounding of RCU.

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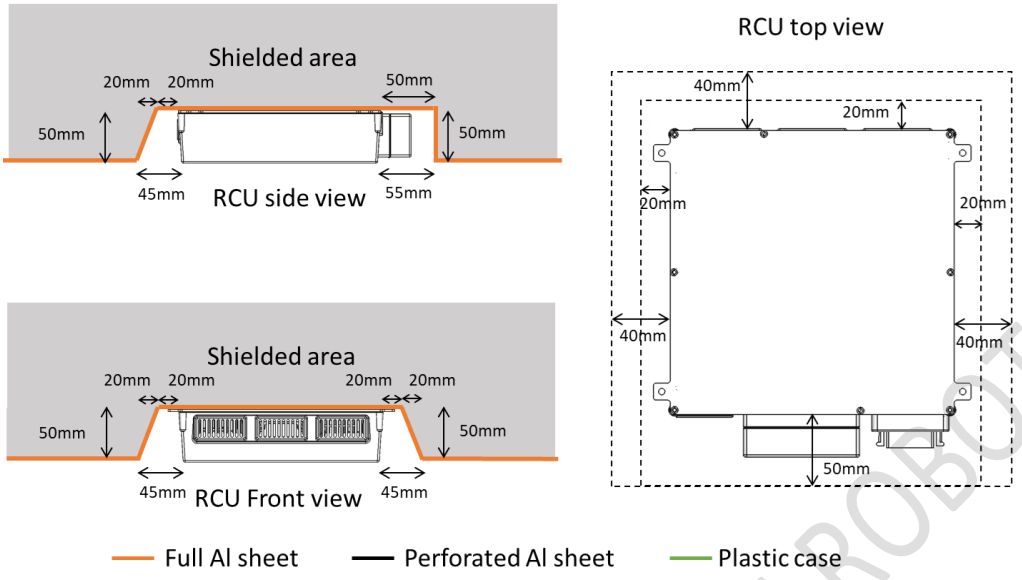

Figure 5: Shielding distance for flushed exposed RCU mounting

Figure 5 details the recommended closest distance for the shielding sheet to the RCU. The electronics can be installed in the shielded area (grey coloured area) behind the metal sheet.

3. Flushed enclosed RCU
   RCU is mounted with flushed surface with the robot and enclosed with plastic casing (non-conductive, non-ferrous material, no carbon fibre) as shown in Figure 5. As the RCU is enclosed in the robots, perforated Al sheets are used at the side of air inflow (grill) and air outflow (fan).

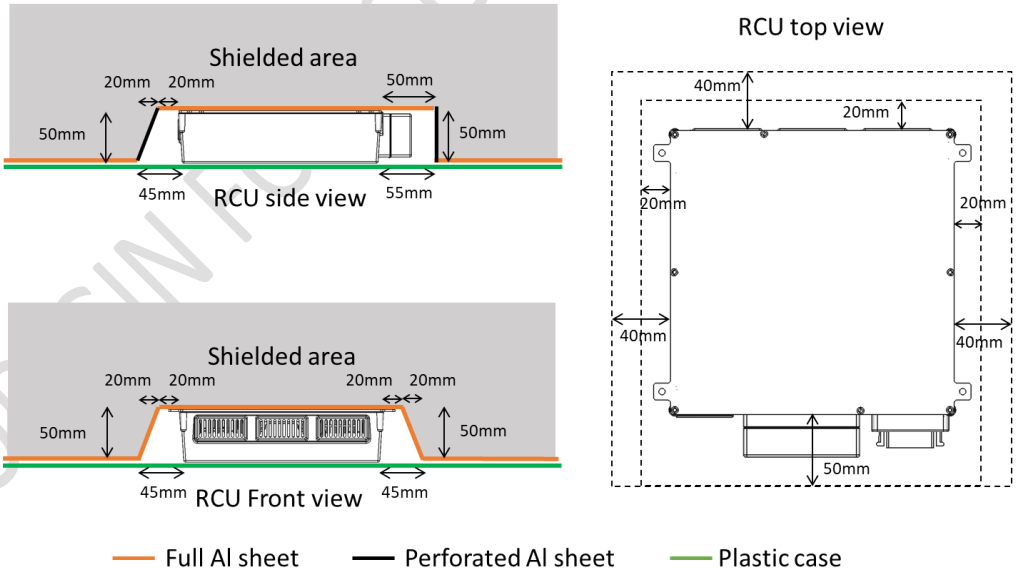

Figure 6: Shielding distance for flushed enclosed RCU mounting

Figure 6 details the recommended closest distance for the shielding sheet to the RCU. The electronics can be installed in the shielded area (grey coloured area) behind the metal sheet.

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Note that installation of shielding sheet at closer distance to RCU may results in high temperature in shielding sheet. On the other hand, installation of shielding sheet at further distance to RCU is preferred and recommended. Furthermore, installation of the plastic case (green line) must include minimum distance of 2 mm from RCU surface. This is to separate hot surface of RCU from the plastic case.

Figure 7 shows the tested Al plate shielding according to dimension given in Figure 5. Figure 8 and Figure 9 shows the shield distance at the air inflow and air outflow side accordingly.

Figure 7: Side view of tested Al plate shielding

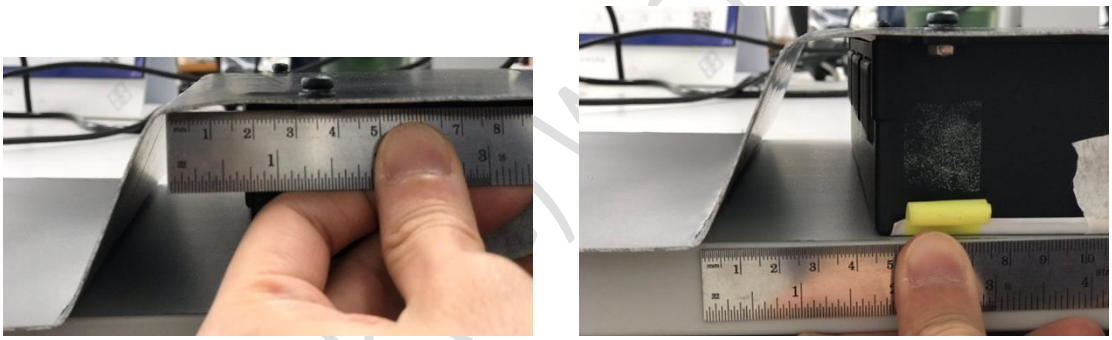

Figure 8: Side view of Al plate shielding at the air inflow side (grill)

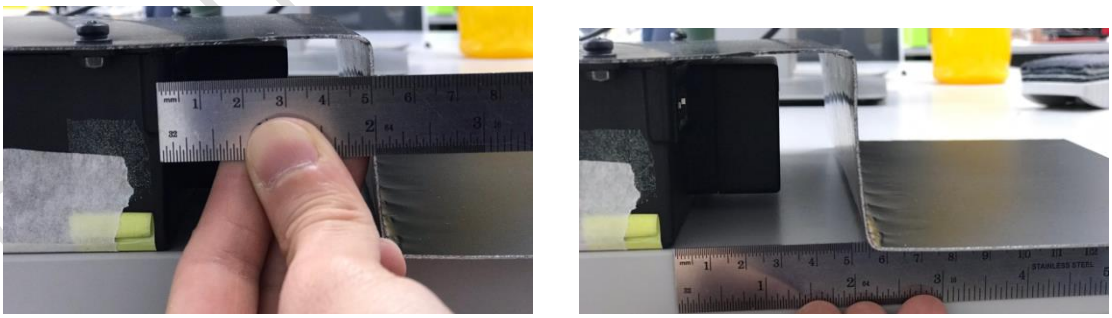

Figure 9: Side view of Al plate shielding at the air outflow side (fan)

Xnergy provides simple RCU shielding as shown in Figure 10 with clearance distance of 20 mm, 30 mm, and 40 mm. The temperature increase in RCU shield ranges from 10^0 C (40 mm clearance) to 25^0 C (20 mm clearance). Please contact Xnergy sales and support if you require this part.

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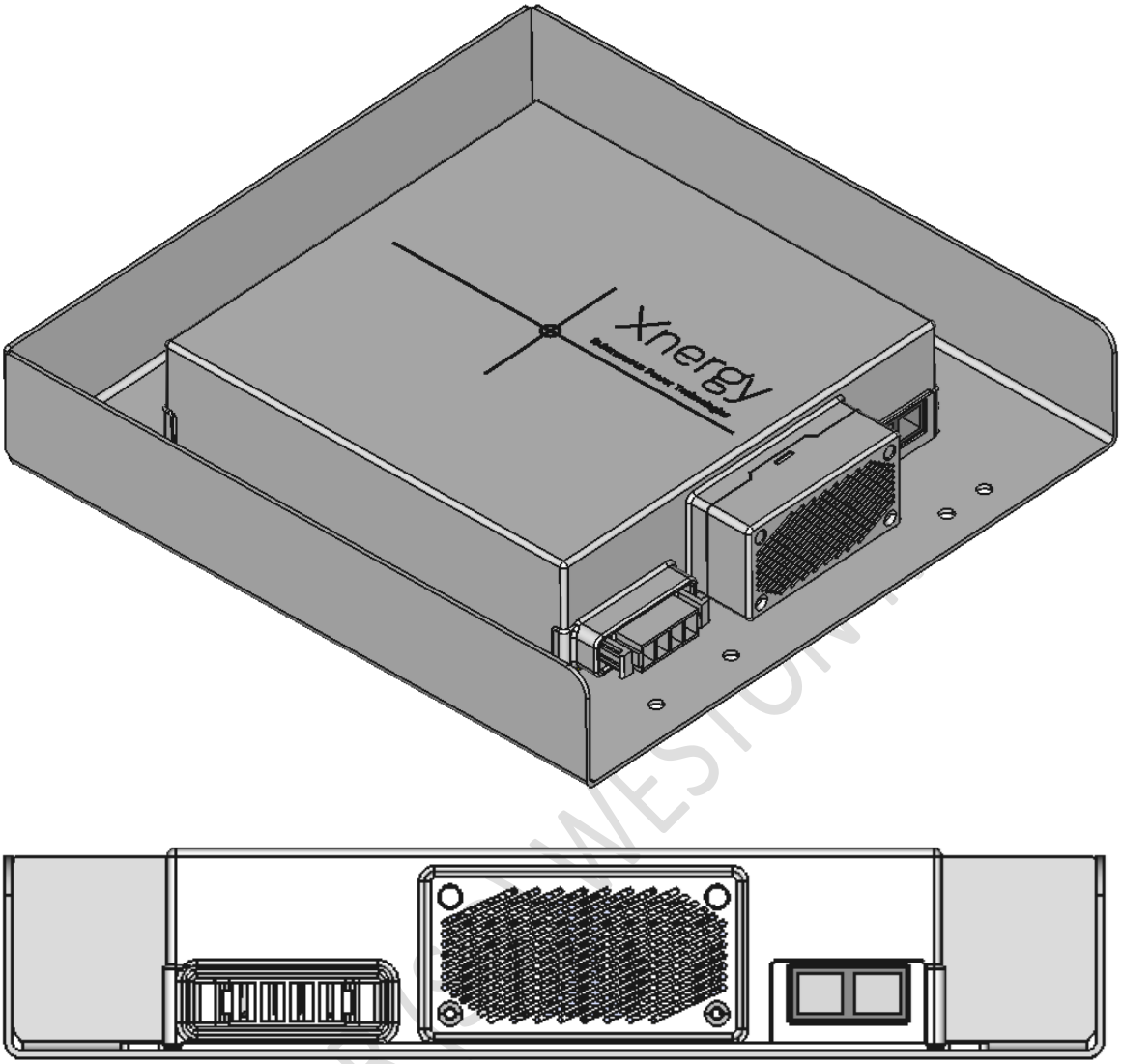

Figure 10: RCU shielding with 40mm clearance

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Shielding distance to TXP

The shielding distance guidelines for TXP are set based on the same principles.

1. Protruding exposed TXP
   TXP is mounted with protruding surface. The electronics can be installed in the shielded area (grey coloured area) behind the full Aluminium sheet (do not use steel).

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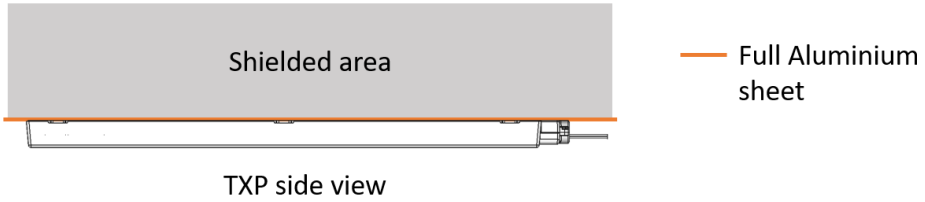

Figure 11: Shielding distance for protruding exposed TXP mounting

2. Flushed TXP

For flushed TXP installation, it can be mounted with exposed TXP or enclosed TXP as shown in Figure 12 and Figure 13. In both cases, the electronics can be installed in the shielded area (grey coloured area) behind the full Aluminium sheet (do not use steel).

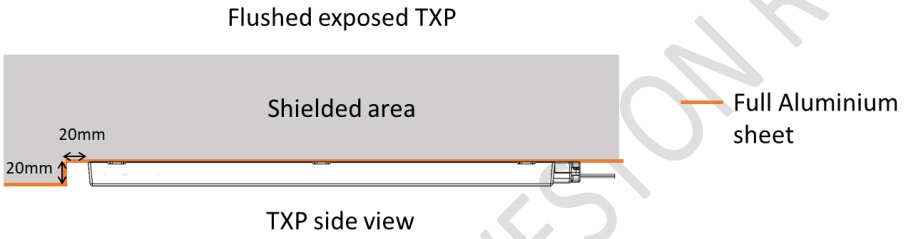

Figure 12: Shielding distance for flushed exposed TXP mounting

The plastic enclosure casing should be made of non-conductive material (no carbon fibre). Furthermore, installation of the plastic case (green line) must include minimum distance of 2 mm from TXP surface. This is to separate hot surface of TXP from the plastic case.

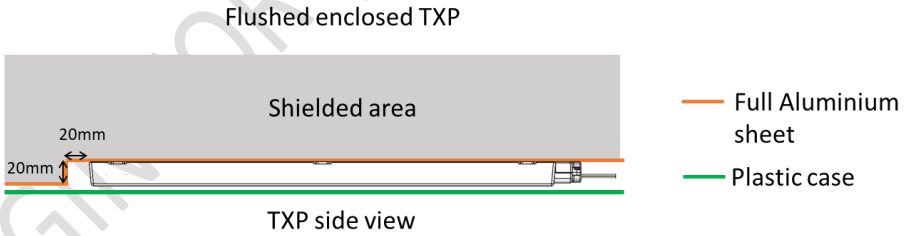

Figure 13: Shielding distance for flushed enclosed TXP mounting

Xnergy provides TXP shielding as shown in Figure 14 with clearance distance of 20 mm. The temperature increase in TXP shielding is up to 25^0 C. Please contact Xnergy sales and support if you require this part.

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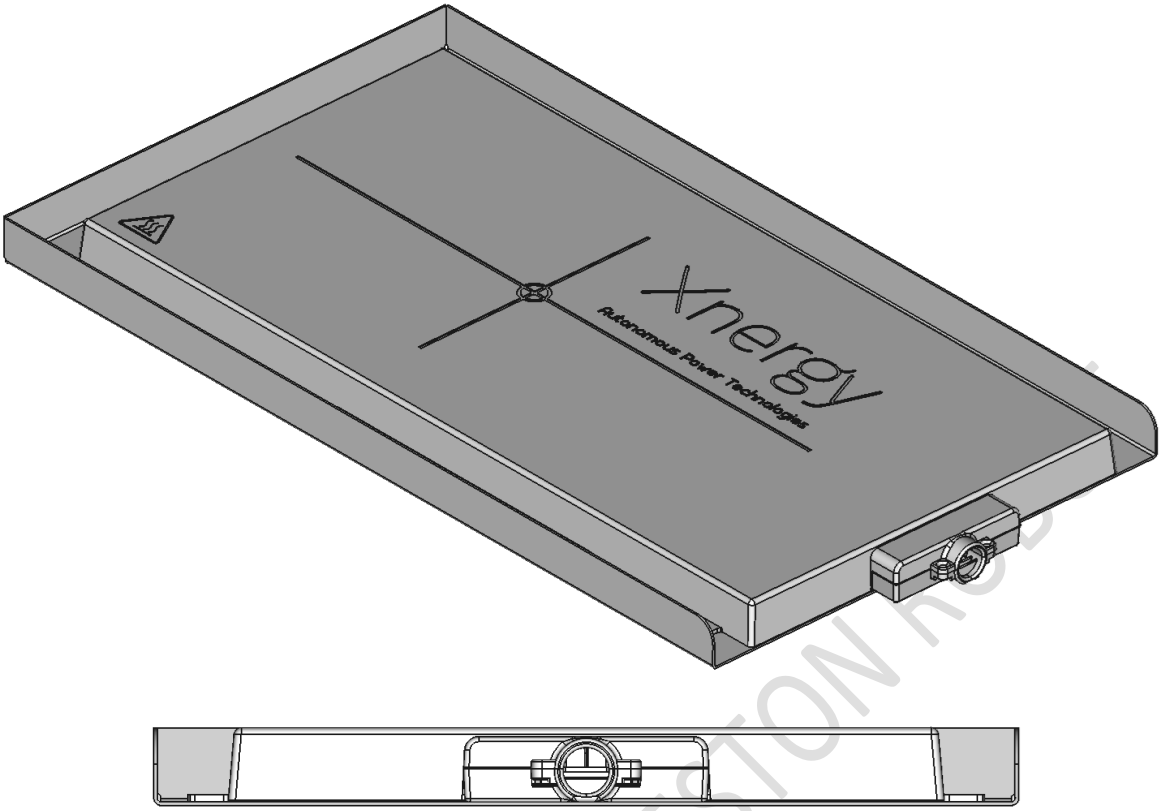

Figure 14: TXP shielding with 20mm clearance

4. Summary

In summary, electronics placed around the TXP and RCU must be shielded to ensure proper operations. The recommended shielding are:

- Aluminium sheet with thickness ≥ 1mm.
- In protruding RCU mounting, the shield is placed behind the RCU's metal plate.
- In flushed RCU mounting, the shield surrounding RCU is placed with minimum clearance distance of 2 cm from RCU metal plate and 4.5 cm from RCU surface (refer to Figure 5 and Figure 6).
- When needed, perforated Aluminium shield with holes of diameter ≤ 3mm can be used at the side of air inflow (grill) and air outflow (fan).
- For RCU shielding,

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5. References

[1] Digikey, "Infineon Products for Service Robotics and Automated Guided Vehicles (AGVs)," [Online]. Available: https://www.digikey.sg/en/product-highlight/i/infineon/service-robotics-and-automated-guided-vehicles?utm_adgroup=&utm_source=google&utm_medium=cpc&utm_campaign=Dynamic%20Search_EN_Product&utm_term=&productid=&gclid=Cj0KCQjw59n8BRD2ARIsAAmgPmK6mOjchsaMRvA.

[2] European Committee for Electrotechnical Standardization (CENELEC) , "Investigation Results on Electromagnetic Interference in the Frequency Range below 150 kHz," Brussels, Belgium, 2017.

6. Revision History

Version control			
Version Number	**Date issued**	**Author**	**Update information**
V1.0	2021.01.21	Dr. Arie Nawawi	First published version
V2.0	2021.03.25	Dr. Arie Nawawi	Updated for TXP

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